

Computational Viability Certification under Incomplete Observation

Supplementary Material: Complete Proofs and the Verification Record

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S1. Complete proofs

Throughout, labels are $a = (\theta, i, t) \in \mathcal{S} = \Theta \times \{1, \dots, s\} \times [0, T]$, k_a , β_a , $\mathcal{F}_a$, $J_{\mathcal{I}}(T)$, ϕ_U , $\Gamma_{\mathcal{I}}$, $\rho_{\mathcal{H},G}$, and the program (5) are as in the main text.

Proof of Proposition 1 (robust rows). The linear solution formula gives, for $t \leq T$,

$$c_i^\top x_\theta(t) = c_i^\top X_\theta(t, 0)x_0 + \int_0^t c_i^\top X_\theta(t, \sigma)g_\theta(\sigma) d\sigma + \int_0^t k_a(\sigma)^\top u(\sigma) d\sigma + \int_0^t c_i^\top X_\theta(t, \sigma)E_\theta(\sigma)d(\sigma) d\sigma.$$

The initial-state term is bounded over $x_0 \in P_\theta$ by the support value $h_{P_\theta}(X_\theta(t, 0)^\top c_i)$, and the disturbance term, maximized over all measurable $d(\cdot)$ with values in D_θ , is $\int_0^t h_{D_\theta}(E_\theta(\sigma)^\top X_\theta(t, \sigma)^\top c_i) d\sigma$; for polytopic D_θ a maximizing vertex selection is measurable, so the bound is attained by an admissible disturbance signal. Collecting terms, the constraint $c_i^\top x_\theta(t) \leq b_i$ holds for every (x_0, d) if and only if $\mathcal{F}_a(\pi) \leq 0$, and the separation of the maximizations from the control term is exactly that the control signal in a mode does not depend on x_0 or d .

Proof of Proposition 2 (value and nonviability). The admissible signal space is a finite product of the sets $L^\infty(I_r; U)$; because U is compact and convex these are weak-* compact by the Banach–Alaoglu theorem. The safety functionals are affine and weak-* continuous because their kernels k_a lie in L^1 . Their supremum over the finitely many labels is therefore weak-* lower semicontinuous and attains its minimum on the policy space: $J_{\mathcal{I}}(T)$ is well defined. Safety of some policy is $\max_a \mathcal{F}_a(\pi) \leq 0$, so $B_0 \in \mathcal{K}_{\mathcal{I}}^T$ iff $J_{\mathcal{I}}(T) \leq 0$; since safety at the margin is viability (constraints closed) and $J_{\mathcal{I}}(T) = 0$ admits safe policies, nonviability is $J_{\mathcal{I}}(T) > 0$. The strict inequality is thus a consequence of compactness and closedness, not an added assumption.

Proof of Theorem 1. Step 1 (control-moment inequality). For a block J_j and any measurable u with values in U ,

$$\left(\int_{J_j} k_a \right)^\top \frac{1}{|J_j|} \int_{J_j} u - \int_{J_j} k_a^\top u = \int_{J_j} (\bar{k}_{a,\mathcal{H}} - k_a)^\top u \leq \int_{J_j} h_U(\bar{k}_{a,\mathcal{H}} - k_a),$$

using $\sup_{v \in U} w^\top v = h_U(w)$. Summing over blocks proves (3); the averages remain compatible with the information structure because the mesh contains all sensing times, so common-history controls use the same block variable. For the bound, choose a center u_c of U ; since $\int_{J_j} (\bar{k} - k) = 0$, the center cancels and $e_{a,\mathcal{H}} \leq R_U \|k_a - \bar{k}_{a,\mathcal{H}}\|_{L^1}$. A bounded-variation function satisfies $\|k_a - \bar{k}_{a,\mathcal{H}}\|_{L^1} \leq h_u \text{TV}(k_a)$, whence $e_{a,\mathcal{H}} \leq R_U V_U h_u$.

Step 2 (lower bound of (8)). Take any admissible continuous-time policy and form its block averages, which lie in U and satisfy (3). For every grid label,

$$A_a v - \beta_a^+ - \bar{e}_a \leq \int_0^T k_a^\top u_\theta^\pi - \beta_a = \mathcal{F}_a(\pi),$$

so the finite program attains a value no larger than the maximum robust violation of that policy. Minimizing over policies gives $\rho_{\mathcal{H},G} \leq J_{\mathcal{I}}(T)$.

Step 3 (upper bound of (8)). Take an optimizer v^* of (5) and implement the corresponding piecewise-constant information-adapted control, for which $\int_0^T k_a^\top u_\theta^{v^*} = A_a v^*$. At grid times, $\mathcal{F}_a(v^*) \leq \rho_{\mathcal{H},G} + \bar{e}_a + \delta_\beta$; the Lipschitz hypothesis (6) extends the bound to every time, giving $\max_a \mathcal{F}_a(v^*) \leq \rho_{\mathcal{H},G} + R_U V_U h_u + \delta_\beta + L h_t$. Since $J_{\mathcal{I}}(T)$ minimizes over all admissible policies, the upper bound follows. Together with (4) this gives the sharpened form. The asymmetry — averaging arbitrary controls proves the lower bound, implementing piecewise-constant controls proves the upper bound — is what makes naïve infeasibility-by-discretization arguments unsound.

Step 4 (sound finite certificates). Write the block input constraints as $F_{\text{blk}} v \leq f_{\text{blk}}$. The program’s dual is

$$\max_{\lambda, \mu} -\lambda^\top (\beta^+ + \bar{e}) - \mu^\top f_{\text{blk}} \quad \text{s.t.} \quad A^\top \lambda + F_{\text{blk}}^\top \mu = 0, \quad \lambda \geq 0, \quad \mathbf{1}^\top \lambda = 1, \quad \mu \geq 0, \quad (12)$$

so a positive dual objective certifies infeasibility of $\{Av \leq \beta^+ + \bar{e}, v \in U^Q\}$. By Step 1 every genuinely safe measurable controller would generate a feasible v ; the finite infeasibility certificate therefore excludes every such controller. Moreover the weighted continuous-time violation is at least the dual objective: a normalized certificate with margin $\gamma > 0$ guarantees an actual facet violation of at least γ in one stored test (Remark 1 of the main text).

Step 5 (completeness and exact duality). If the system is nonviable, let $\gamma = J_{\mathcal{I}}(T) > 0$ and choose meshes making the error in (8) smaller than γ ; then $\rho_{\mathcal{H},G} > 0$ and finite LP duality yields a positive certificate, proving (iii). Every finite atomic choice satisfies $\Gamma_{\mathcal{I}}(\lambda, a) \leq J_{\mathcal{I}}(T)$ (the same averaging computation as in Step 1 applied to the pooled kernels), and an optimal dual solution of (5) yields an atomic continuous-time lower bound at least $\rho_{\mathcal{H},G}$; letting the mesh error tend to zero proves (9). No belief-state recursion is used.

Step 6 (sparsity). Factor the finite row map through its rank, $Av = L(Rv)$, $Rv \in \mathbb{R}^r$; the attainable moment set $\mathcal{M} = \{Rv : v \in U^Q\}$ is compact and convex, and each safety row defines a halfspace in $\mathbb{R}^r$. If the common feasible set is empty, Helly’s theorem in $\mathbb{R}^r$ gives an obstruction involving at most $r + 1$ safety halfspaces, and finite duality on that subproblem yields a sparse positive certificate with the same objective: $r + 1 \leq mQ + 1$ labels. To preserve a prescribed margin $\gamma_0 < \rho_{\mathcal{H},G}$, apply the argument to the still-infeasible shifted constraints $A_a v \leq \beta_a^+ + \bar{e}_a + \gamma_0$. The continuous finite-rank case follows identically on the compact attainable moment body of the lifted kernels, giving $r_{\mathcal{I}} + 1$. $\square$

Proof of Proposition 3 (arbitrarily large minimal obstructions). With the construction of the main text and $w_j = \int_{I_j} u$: mode $j \leq q$ ends the interval at $x_j = w_j - 1 \geq 0$, so safety of the blind window requires $w_j \geq 1$; mode $q + 1$ ends at $q - \varepsilon - \sum_j w_j$, so safety requires $\sum_j w_j \leq q - \varepsilon$. Summing the q lower bounds against the upper bound gives $0 \leq -\varepsilon$: a contradiction, so the full belief is nonviable. Removing mode $q + 1$, $u \equiv 1$ gives $w_j = 1$ and $x_j = 0$: safe. Removing mode $j \leq q$, $u = 0$ on I_j and $u = 1$ elsewhere gives $x_{q+1} = (q - 1) - (q - \varepsilon) + 1 = \varepsilon \geq 0$ at the end and safety on the remaining branch: every proper subbelief is viable. Equal weights $\lambda_j = 1/(q+1)$ on the $q + 1$ endpoint rows give pooled pre-kernels $\sum_j \lambda_j k_j = 0$ on every interval where the modes agree and a margin of $\varepsilon/(q+1)$ at the horizon. The kernel rank: the endpoint rows are $\mathbf{1}_{I_j}$ -weighted copies of the single input direction with one linear relation (their weighted sum), so the rank of the safety-row matrix is exactly q , and Helly-type sparsity in that rank requires $q + 1$ witnesses — no bound in the input dimension $m = 1$ exists. $\square$

Proof of the erosion identity. ϕ_U is concave and positively homogeneous; for any decomposition, $\phi_U(\sum_j g_j) \geq \sum_j \phi_U(g_j)$ because each $\phi_U(g_j) = \min_{v \in U} g_j^\top v$ is attained at a common minimizer for the sum while separate minima only lower the total. Consequently the same witness family, evaluated on the coarse partition, dominates its evaluation on the refinement, $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$, and the exact erosion is the displayed integral (12). For $U = [-\bar{u}, \bar{u}]^m$, $\phi_U(g) = -\bar{u}\|g\|_1$, giving the stated ℓ^1 form. The 3,240-pair verification for the hexagonal U is reported in S2. $\square$

Proof of the oracle-recourse certificate (13). The oracle relaxation replaces the true post- τ policy class by the larger class that may assign each stored scenario its own measurable control. The certificate (13) is the atomic margin (1) evaluated in that enlarged class: before τ the pooled kernel term is $\phi_U(\sum_\ell \lambda_\ell k_\ell)$, after τ each scenario ℓ contributes $\phi_U(\lambda_\ell k_\ell)$ with its own margin β_ℓ computed from the stored scenario. A positive margin certifies infeasibility in the enlarged class by Propositions 1–2; every policy of the true class is feasible for the enlarged one, so the verdict transfers downward: no actual output-feedback policy is safe. Soundness is thus immediate; completeness fails because the oracle class is strictly larger.

S2. The verification record

All numbers in the main text are re-derived in exact rational arithmetic (Python `fractions`; no floating point in any witness computation; floats only inside the solver call, whose output is compared against the exact value only within solver tolerance). Two scripts carry the record, both at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`):

- `paper2c_lp_campaign.py` — the solver-certified mesh campaign: builds program (5) at each mesh $h \in \{0.2, 0.1, 0.05, 0.02, 0.01\}$, verifies the dimension identities ($mQ+1$ variables; $Ms(N_t+1) + p_U Q$ rows), solves with HiGHS, and certifies optimality of $\rho(h) = \max\{0, \frac{3}{50} - Th/4\}$ by the exact primal witness

(pre-observation blocks 0; post-observation blocks of mode j at $-n_j$; every row checked) and the exact dual witness ($\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$, $\mu_{j,k} = \lambda_j a_k \eta_j$; stationarity and objective $-\rho(h)$ per block). Also re-derives $\Gamma(\tau) = \tau - \frac{7}{50}$, $\tau_{\max} = \frac{7}{50}$, the $h = \frac{1}{10}$ Farkas contradiction $-\frac{3}{100}$, and the pair-viability positions 1.9752 and 1.9352. Current run: 9/9 checks pass; solver agreement at most 4.2×10^{-17} on all five meshes.

- **paper2_nonstandard_verification.py** — the 51-check exact-arithmetic record for the predecessor material: the three-branch instance values, the $m + 1$ construction for all tested q , the ϕ_U superadditivity on 3,240 rational pairs, the error-bound checks, and the counterexamples. Current run: 51/51 checks pass.

The mesh table of the main text is regenerated verbatim by the first script; the reported iteration counts and wall-clock times are measured values from the run recorded here (scipy HiGHS, dual simplex), not estimates.